ST. JOSEPH COLLEGE OF ENGINEERING AND TECHNOLOGY
NTA LEVEL 5 - TECHNICIAN CERTIFICATE
CONTINUOUS ASSESSMENT TEST - II - MARCH 2026
EDT 05102 - ENGINEERING MATHEMATICS

SEMESTER: I — **YEAR: II**
DURATION: 1 Hr 30 Min. — **MAX. MARKS: 50**

PART – A (Answer all the Questions) (7 x 2 = 14)

1. Evaluate $\int \cot^2 x dx$
2. Define Calculus.
3. What is conjugate of complex number?
4. Write the properties of imaginary unit.
5. Define the Demoivre's theorem.
6. Find the cube roots of unity $z^3 = 1$
7. Write the applications of complex numbers.

PART – B (Answer any three Questions) (3 x 4 = 12)

8. Differentiate: $y = x^2 \sin x, \quad y = \dfrac{x^2}{\sin x}$.
9. Differentiation of $\tan x$ from the first principles.
10. Prove that $\cos 15^\circ - \sin 15^\circ = \dfrac{1}{\sqrt{2}}$
11. Prove that $\dfrac{\sin 75^\circ - \sin 15^\circ}{\cos 75^\circ + \cos 15^\circ} = \dfrac{1}{\sqrt{3}}$
12. Find the middle term in the expansion of $\left(2x - \dfrac{3}{x}\right)^5$
13. Find the value of $(10.1)^5$ using Binomial theorem.

PART – C (Answer any two Questions) (2 x 12 = 24)

14. Find the volume generated when the area bounded by $y^2 = 25x^3$ revolves about the x-axis between x=2 and x=3.
15. (i) Derive the formula for the division of two complex numbers.
(ii) Derive the formula for the multiplication of two complex numbers.
16. Find $z_1 + z_2,\ z_1 \times z_2$ for the followings:
(i) $z_1 = 6 + 4i, z_2 = 2 - 7i$,
(ii) $z_1 = 9 - 3i, z_2 = 7 + 8i$
17. (i) Discuss the applications of complex numbers in engineering.
(ii) How can an electrical engineering problem involving impedance be solved using complex numbers?
(iii) Illustrate the use of complex numbers in solving a mechanical engineering vibration problem.